

GRANT FLORES-DEWILDE

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EDUCATION

University of Pennsylvania, Ben Franklin Scholar | *Philadelphia, PA*
BSE in Mechanical Engineering | Expected Graduation May 2029

3.82/4.0 GPA

PROFESSIONAL EXPERIENCE

Penn Electric Racing (FSAE) | *Philadelphia, PA* | Mechanical Team - Accumulator Designer

Sept. 2025 - Present

- Responsible Engineer for High-Voltage Tractive Battery Pack & Battery Junction Box; both CADed, manufacturing in progress. HV Pack is mainly welded aluminum panels, while Junction Box is 3D printed from Ultem.
- Led design of new custom Vibration Table to stress-test battery pack at 334% higher theoretical excitation frequency vibration than previous years, while increasing stability and allowing for axial control through linear bearing restriction.
- Designed new custom Charge Cart for battery pack, fabricated and adjusted enclosure sizing based on electrical subteam's needs.
- Designed Suspension Chassis Tabs and Evaluated Safety Utilizing Finite Element Analysis (FEA).
- Developed and Manufactured 14 Firewall Jigs and Rear Tire Shock Master Jig.

ModLab | *Pennovation Works* | Undergrad Researcher

May 2026 - Present

- Designed and fabricated fatigue testing structure for twisted string actuator in the ABOT project, a self-configurable robot composed of collapsible linkages, optimized actuator and testing rig design based on fatigue testing results.

Precision Machining Laboratory | *University of Pennsylvania MEAM Labs* | Machine Lab Assistant

August 2026 - Present

- Oversee machine shop, provide machining advice and demonstrations to students when needed
- Ensure student safety and keep machines clean

After-The-Bell | *Kennett Middle School* | Club Founder & President

August 2021 - May 2025

- Organized high school volunteers, ran fundraising events, helped with tutor and other activities, led robotics sessions, conducted a high school focus group, for a free after-school program that serves over 300 students.

FIRST Tech Challenge (Robotics Competition) | Captain & Driver

May 2022 - May 2025

- Driver at competitions, led Outreach and Sponsorship subteams, ran community outreach, robotics classes, and technology demos at schools, libraries, and museums.

PROJECTS

ABOT Fatigue Testing, University of Pennsylvania ModLab | *Philadelphia, PA*

Summer 2026-Present

- Built a structure to test universal twisted string actuator lifecycle
- 500% increase in lifecycle from my design, from 10K to 50K actuated cycles
- Results submitted in paper to IEEE ICRA 2027 Conference; received acknowledgement for technical implementation

Stirling Engine, University of Pennsylvania | *Philadelphia, PA*

Spring 2026

- Designed and manufactured custom Spider-Man themed Stirling heat engine
- Learned precision machining techniques, machined all parts within given tolerances
- Tested at 500 RPM
- Hired by class instructor as a machine lab assistant after completing the project

Anchored Compass, University of Pennsylvania | *Philadelphia, PA*

Spring 2026

- Created compass that could always point back to a droppable invisible waypoint with a button press,
- Utilized Arduino, GPS, LED Matrix, and an IMU

Siege Machine, University of Pennsylvania | *Philadelphia, PA*

Fall 2025

- Designed and manufactured a competition catapult to launch 1-inch rubber projectiles with repeatable accuracy across multiple preset launch angles

Analyzing Drag Coefficients of 3D Shapes at Terminal Velocity, PGSS, CMU | *Pittsburgh, PA*

Summer 2024

- Built a wind tunnel and researched the effects of wing shape on drag.
- Led data analysis and error discussion, verified corresponding drag coefficients within 97% of NASA's accepted values.

SKILLS

Software: Python, Java, OCaml, SolidWorks & Fusion 360 CAD, Finite Element Analysis (FEA), Arduino IDE

Hardware: Milling, Laser-cutting, 3D printing, GD&T, Tolerance Stack-up, Circuits

Languages: Spanish, intermediate proficiency